AN IMPROVED LOWER BOUND FOR THE ERDŐS–GINZBURG–ZIV CONSTANT OF RANK-THREE ELEMENTARY p-GROUPS

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Abstract. For an odd prime p, let $\eta(C_p^3)$ denote the smallest integer ℓ such that every sequence of ℓ elements in $(\mathbb{Z}/p\mathbb{Z})^3$ contains a zero-sum subsequence of length at most p. We prove that $\eta(C_p^3) \geq 8(p-1)+1$, improving the classical lower bound of $7(p-1)+1$ due to Harborth. The construction uses eight explicitly given nonzero vectors in $\mathbb{F}_p^3$, and the proof that no short zero-sum exists reduces to an integer inequality established by case analysis on the structure of the kernel. A machine-checked formalization of the proof in Lean 4 accompanies this paper as ancillary material.

1. Introduction

Let G be a finite abelian group with exponent $\exp(G)$. A *sequence* over G is a finite unordered collection of elements of G, with repetition allowed. A nonempty subsequence T of a sequence S is a *zero-sum subsequence* if its elements sum to 0 in G; it is *short* if $|T| \leq \exp(G)$.

Definition 1. The *η-constant* $\eta(G)$ is the smallest positive integer ℓ such that every sequence over G of length ℓ contains a short zero-sum subsequence.

Definition 2. The *s-constant* $\mathsf{s}(G)$ is the smallest positive integer ℓ such that every sequence over G of length ℓ contains a zero-sum subsequence of length *exactly* $\exp(G)$.

Since any zero-sum of length exactly $\exp(G)$ is in particular a short zero-sum, we have $\eta(G) \leq \mathsf{s}(G)$. The classical theorem of Erdős, Ginzburg, and Ziv [2] establishes $\mathsf{s}(C_n) = 2n-1$.

For the elementary p-group $G = C_p^r$ of rank r, a general lower bound follows from the *binary cube construction*: one takes each nonzero element of $\{0,1\}^r \subset \mathbb{F}_p^r$ with multiplicity $p-1$. This sequence of length $(2^r-1)(p-1)$ contains no short zero-sum, giving

$$\eta(C_p^r) \geq (2^r-1)(p-1)+1. \tag{1}$$

This bound, due to Harborth [5], gives $\eta(C_p^3) \geq 7(p-1)+1$ for rank three.

For the s-constant, Elsholtz [3] proved $\mathsf{s}(C_n^3) \geq 9(n-1)+1$ for all odd n, using a nine-element construction that includes the zero vector together with a parity argument. Since the zero vector forms a zero-sum of length 1, his construction is not zero-sum-free for short subsequences; it avoids only zero-sums of length *exactly* n.

Our main result improves the η-bound for rank three:

Theorem 3. *For every odd prime p,*

$$\eta(C_p^3) \geq 8(p-1)+1.$$

The improvement over Harborth's bound is exactly $p-1$: we exhibit eight *nonzero* vectors in $\mathbb{F}_p^3$ such that taking each with multiplicity $p-1$ yields a sequence with no short zero-sum. The proof proceeds by reducing the zero-sum-free property to an integer inequality on the kernel of the associated matrix, then establishing this inequality by case analysis. Parity of p plays a critical

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role in the base case, following the philosophy of Elsholtz's argument for the s-constant. As a consequence, we also recover $\mathsf{s}(C_p^3) \geq 9(p-1)+1$ for odd primes (Corollary 13).

The proof has been formalized in the Lean 4 proof assistant [6] using the Mathlib library, in a self-contained file of 541 lines; the source file `DavenportAll.lean` is included as ancillary material.

2. The construction

Fix an odd prime p. Define eight vectors in $\mathbb{F}_p^3$:

$$\mathcal{A}_8 = \{\mathbf{a}_0, \mathbf{a}_1, \ldots, \mathbf{a}_7\}, \tag{2}$$

where

$$\begin{aligned} \mathbf{a}_0 &= (1,\ 0,\ 0), & \mathbf{a}_4 &= (1,\ 1,\ 0),\\ \mathbf{a}_1 &= (0,\ 1,\ 0), & \mathbf{a}_5 &= (1,\ 0,\ 1),\\ \mathbf{a}_2 &= (0,\ 0,\ 1), & \mathbf{a}_6 &= (-1,\ 2,\ 1),\\ \mathbf{a}_3 &= (-1,\ 1,\ 1), & \mathbf{a}_7 &= (-1,\ 1,\ 2). \end{aligned} \tag{3}$$

All eight vectors are nonzero in $\mathbb{F}_p^3$ for every prime $p \geq 3$. Five of them—$\mathbf{a}_0, \mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_4, \mathbf{a}_5$—belong to the binary cube $\{0,1\}^3$; the remaining three lie outside it.

Define the sequence

$$S = \prod_{j=0}^{7} \mathbf{a}_j^{p-1}, \tag{4}$$

so $|S| = 8(p-1)$. Theorem 3 is equivalent to the assertion that S contains no zero-sum subsequence of length at most p.

Remark 4. The binary cube $\{0,1\}^3 \setminus \{0\}$ has seven elements. Harborth's construction uses all seven with multiplicity $p-1$. Going beyond seven nonzero vectors requires using elements outside $\{0,1\}^3$; the specific choice (3) was found by computational search and verified for $p = 3, 5, 7, 11, 13$.

3. Kernel parametrization

Arrange the eight vectors as columns of a 3×8 matrix:

$$M = \begin{pmatrix} 1 & 0 & 0 & -1 & 1 & 1 & -1 & -1 \\ 0 & 1 & 0 & 1 & 1 & 0 & 2 & 1 \\ 0 & 0 & 1 & 1 & 0 & 1 & 1 & 2 \end{pmatrix}. \tag{5}$$

The first three columns form the identity, so $\operatorname{rank}(M) = 3$ over $\mathbb{F}_p$ for every prime $p \geq 3$.

A subsequence of S selects multiplicities $c_j \in \{0, 1, \ldots, p-1\}$ for each $\mathbf{a}_j$. It sums to zero in $\mathbb{F}_p^3$ if and only if $M\mathbf{c} \equiv \mathbf{0} \pmod{p}$, where $\mathbf{c} = (c_0, \ldots, c_7)^T$. The kernel $\ker(M)$ over $\mathbb{F}_p$ has dimension 5.

We parametrize $\ker(M)$ by the five free coordinates

$$(t_0, t_1, t_2, t_3, t_4) := (c_3, c_4, c_5, c_6, c_7),$$

each in $\{0, 1, \ldots, p-1\}$. Solving $M\mathbf{c} \equiv \mathbf{0}$ for (c_0, c_1, c_2) gives:

$$c_1 \equiv -A(t) \pmod{p}, \qquad A(t) := t_0 + t_1 + 2t_3 + t_4, \tag{6}$$

$$c_2 \equiv -B(t) \pmod{p}, \qquad B(t) := t_0 + t_2 + t_3 + 2t_4, \tag{7}$$

$$c_0 \equiv C(t) \pmod{p}, \qquad C(t) := t_0 - t_1 - t_2 + t_3 + t_4. \tag{8}$$

Definition 5. For $t = (t_0, \ldots, t_4) \in \{0, \ldots, p-1\}^5$, let $c_j = c_j(t)$ be the unique representative in $\{0, \ldots, p-1\}$ satisfying the congruences above, and define the *lift-weight*

$$w(t) := \sum_{j=0}^{7} c_j(t) = (t_0 + t_1 + t_2 + t_3 + t_4) + c_0(t) + c_1(t) + c_2(t).$$

The lift-weight equals the length of the subsequence of S corresponding to $\mathbf{c}(t)$. Therefore S is free of short zero-sums if and only if $w(t) \geq p+1$ for every nonzero t.

Proposition 6 (Main inequality)**.** *For every odd prime $p \geq 3$ and every nonzero $t \in \{0, \ldots, p-1\}^5$,*

$$w(t) \geq p+1.$$

The remainder of the paper proves Proposition 6.

4. The weight formula

Writing c_j in terms of the modular reductions introduces integer *wrap counts* $n_0, n_1, n_2 \in \mathbb{Z}$ defined by

$$c_1 = pn_1 - A, \quad c_2 = pn_2 - B, \quad c_0 = C + pn_0, \tag{9}$$

where the conditions $0 \leq c_j \leq p-1$ constrain (n_0, n_1, n_2) uniquely. Write $N := n_0 + n_1 + n_2$ for the *total wrap count*.

We use the following auxiliary quantities throughout the proof:

$$T := t_3 + t_4, \qquad R := t_0 + T, \qquad S_4 := t_1 + t_2 + t_3 + t_4. \tag{10}$$

Elementary bounds: $0 \leq T \leq 2(p-1)$, $0 \leq R \leq 3(p-1)$, $0 \leq S_4 \leq 4(p-1)$. The following identities are verified by direct calculation:

$$A = R + t_1 + t_3, \quad B = R + t_2 + t_4, \quad C = R - (t_1 + t_2), \quad A + B = 2R + S_4. \tag{11}$$

In particular, $A \geq S_4 - (p-1)$ and $B \geq S_4 - (p-1)$ (using $t_0 \geq 0$ and the bound on the negative term).

Lemma 7 (Weight identity)**.** $w(t) = pN - S_4$.

Proof. Expanding $w = (t_0 + S_4) + (pn_1 - A) + (pn_2 - B) + (C + pn_0)$ and applying (11) yields $w = pN - S_4$. □

Thus Proposition 6 is equivalent to $pN - S_4 \geq p+1$, i.e., $S_4 \leq p(N-1) - 1$, for all nonzero t.

5. Proof of the main inequality

5.1. Step 1: $N \geq 2$.

Lemma 8. *For every nonzero t with valid wrap counts, $N \geq 2$.*

Proof. Since $A, B \geq 0$ and $c_1, c_2 \geq 0$, we have $n_1, n_2 \geq 0$. Moreover, $A > 0$ implies $n_1 \geq 1$ (since $0 \leq pn_1 - A < p$ forces $n_1 \geq 1$), and similarly for n_2.

Case $t_0 = 0$, $T = 0$: Then $t_3 = t_4 = 0$, $A = t_1$, $B = t_2$, $C = -(t_1 + t_2)$. Since $t \neq 0$, at least one of t_1, t_2 is positive, so $C \leq -1$, forcing $n_0 \geq 1$; the positive t_j gives the corresponding $n_j \geq 1$, so $N \geq 2$.

Case $t_0 = 0$, $T \geq 1$: Then $R = T \geq 1$, so $A, B > 0$ and $n_1, n_2 \geq 1$. If $C < p$, then $n_0 \geq 0$ and $N \geq 2$. If $C \geq p$, then $T \geq p$ (since $C = T - (t_1 + t_2) \geq p$ and $t_1, t_2 \geq 0$), forcing $t_3, t_4 \geq 1$ (as each is at most $p-1$ and $t_3 + t_4 \geq p$). Then $A = T + t_1 + t_3 \geq p+1$ and $n_1 \geq 2$; similarly $n_2 \geq 2$. Since $C \leq T \leq 2(p-1) < 2p$, we have $n_0 \geq -1$, giving $N \geq -1 + 2 + 2 = 3$.

Case $t_0 \geq 1$: Then $R \geq 1$, so $A, B > 0$ and $n_1, n_2 \geq 1$. If $n_0 \geq 0$ we are done. If $n_0 = -1$ and $n_1 + n_2 \geq 3$, then $N \geq 2$. If $n_0 = -1$ and $n_1 = n_2 = 1$, then $A, B \leq p$ and $C \geq p$, i.e., $R \geq p + t_1 + t_2$. From $A = R + t_1 + t_3 \leq p$ we get $t_1 + t_3 \leq -(t_1 + t_2)$, forcing $t_1 = t_2 = t_3 = 0$. Similarly $B = R + t_2 + t_4 \leq p$ gives $t_4 = 0$. Then $C = t_0 \leq p - 1 < p$, contradicting $C \geq p$. If $n_0 \leq -2$, then $C \geq 2p$, so $R \geq 2p$ and $T \geq p+1$ (since $t_0 \leq p-1$). Each $t_3, t_4 \geq 2$, giving $A, B \geq 2p+2$ and $n_1, n_2 \geq 3$. One checks $n_0 \geq -2$ (if $n_0 \leq -3$, then $C \geq 3p$, but $C \leq R \leq 3(p-1) < 3p$), so $N \geq -2 + 3 + 3 = 4$. □

5.2. **Step 2:** $N \geq 5$**.** If $N \geq 5$, then $w = pN - S_4 \geq 5p - 4(p-1) = p + 4 \geq p + 1$.

5.3. **Step 3:** $N = 4$**.**

Lemma 9. *If $N = 4$ and $S_4 \geq 3p$, a contradiction arises.*

Proof. From $A, B \geq S_4 - (p-1) \geq 2p + 1$, we get $n_1, n_2 \geq 3$ and hence $n_0 \leq -2$. Then $C \geq 2p$, so $R \geq 2p + t_1 + t_2$. Since $S_4 \geq 3p$, we have $t_1 + t_2 = S_4 - T \geq 3p - 2(p-1) = p + 2$, so $C = R - (t_1 + t_2) \leq 3(p-1) - (p+2) = 2p - 5 < 2p$, contradicting $C \geq 2p$. □

Hence when $N = 4$: $S_4 \leq 3p - 1$ and $w = 4p - S_4 \geq p + 1$.

5.4. **Step 4:** $n_0 \geq -1$ **when** $N \leq 3$**.**

Lemma 10. *If $N \leq 3$ then $n_0 \geq -1$.*

Proof. If $n_0 \leq -2$, then $C \geq 2p$, hence $R \geq 2p$. Since $t_0 \leq p - 1$, we get $T \geq p + 1$, and then $t_3, t_4 \geq 2$. This gives $A \geq R + t_3 \geq 2p + 2$ and $n_1 \geq 3$; similarly $n_2 \geq 3$. Then $N \geq n_0 + 3 + 3$. Since $n_0 \geq -2$ (as shown in the proof of Lemma 8), $N \geq 4$, contradicting $N \leq 3$. □

5.5. **Step 5:** $N = 3$**.**

Lemma 11. *If $N = 3$, $n_0 \geq -1$, and $S_4 \geq 2p$, a contradiction arises.*

Proof. Since $A, B \geq S_4 - (p-1) \geq p + 1 > 0$, we have $n_1, n_2 \geq 1$. With $n_0 \in \{-1, 0, 1\}$ and $N = 3$, the triple (n_0, n_1, n_2) is one of: $(1, 1, 1)$, $(0, 2, 1)$, $(0, 1, 2)$, $(-1, 2, 2)$, $(-1, 1, 3)$, $(-1, 3, 1)$. We treat each in turn.

Triple $(1, 1, 1)$*:* Then $A, B \leq p$, so $A + B = 2R + S_4 \leq 2p$. Since $S_4 \geq 2p$ and $R \geq 0$, we need $R = 0$, hence $t_0 = T = 0$. Then $S_4 = t_1 + t_2 \leq 2(p-1) < 2p$, a contradiction.

Triple $(0, 2, 1)$*:* Then $B \leq p$ and $C \geq 0$. From $C = R - (t_1 + t_2) \geq 0$ and $S_4 = t_1 + t_2 + T$, we get $R \geq S_4 - T$, so $t_0 + 2T \geq S_4 \geq 2p$. But $B = R + t_2 + t_4 \leq p$, so $R + t_4 \leq p$ (since $t_2 \geq 0$); hence $t_0 + 2T = (R + t_4) + t_3 \leq p + t_3 \leq p + (p-1) = 2p - 1$, contradicting $t_0 + 2T \geq 2p$.

Triple $(0, 1, 2)$*:* By the same argument with A and B interchanged (replacing $t_3 \leftrightarrow t_4$, $t_1 \leftrightarrow t_2$).

Triple $(-1, 2, 2)$*:* Then $C \geq p$, i.e., $R \geq p + t_1 + t_2$. If $t_1 + t_2 \geq 1$, then $A + B = 2R + S_4 \leq 4p$ (since $A, B \leq 2p$), giving $S_4 \leq 4p - 2R \leq 4p - 2(p+1) = 2p - 2$, a contradiction. If $t_1 = t_2 = 0$, then $S_4 = T \leq 2(p-1) < 2p$.

Triple $(-1, 1, 3)$*:* Then $C \geq p$ and $A \leq p$. From $R \geq p + t_1 + t_2$ and $A = R + t_1 + t_3 \leq p$, we get $t_1 + t_3 \leq -(t_1 + t_2)$, forcing $t_1 = t_2 = t_3 = 0$. Then $S_4 = t_4 \leq p - 1 < 2p$.

Triple $(-1, 3, 1)$*:* By the same argument with $B \leq p$ in place of $A \leq p$, giving $t_1 = t_2 = t_4 = 0$ and $S_4 = t_3 \leq p - 1 < 2p$. □

Hence when $N = 3$: $S_4 \leq 2p - 1$ and $w = 3p - S_4 \geq p + 1$.

5.6. **Step 6:** $N = 2$ **(parity obstruction).**

Lemma 12. *If p is an odd prime, $N = 2$, $n_0 \geq -1$, and $S_4 \geq p$, a contradiction arises.*

Proof. Since $S_4 \geq p$, we have $A, B \geq S_4 - (p-1) \geq 1 > 0$, so $n_1, n_2 \geq 1$. The triple (n_0, n_1, n_2) is one of $(0, 1, 1)$, $(-1, 2, 1)$, $(-1, 1, 2)$.

Triple $(0, 1, 1)$*:* Then $A, B \leq p$ and $C \geq 0$. From $A + B = 2R + S_4 \leq 2p$ and $S_4 \geq p$, we get $2R \leq p$. From $C = R - (t_1 + t_2) \geq 0$ and $S_4 = t_1 + t_2 + T$, we get $R \geq t_1 + t_2 = S_4 - T$, so $R + T \geq S_4 \geq p$. Combined with $2R \leq p$, we get $2T \geq 2p - 2R \geq p$.

Now $R = t_0 + T$ and $2R \leq p$ give $t_0 \leq p/2 - T \leq 0$. Since $t_0 \geq 0$, we have $t_0 = 0$ and $R = T$. Then $2T \leq p$ (from $2R \leq p$) and $2T \geq p$ (from above), so $2T = p$. But $2T$ is even while p is odd—a contradiction.

Triple $(-1, 2, 1)$*:* Then $C \geq p$ and $B \leq p$. From $R \geq p + t_1 + t_2$ and $B = R + t_2 + t_4 \leq p$, we get $t_2 + t_4 \leq -(t_1 + t_2) \leq 0$, so $t_2 = t_4 = 0$ and then $t_1 = 0$. Thus $S_4 = t_3$, $T = t_3$, and

$R = t_0 + t_3 \geq p$ (since $t_1 = t_2 = 0$). From $A = R + t_3 \leq 2p$ (since $n_1 = 2$) and $R \geq p$, we get $t_3 \leq p$. But $S_4 = t_3 \geq p$ and $t_3 \leq p - 1$ gives $S_4 \leq p - 1 < p$, a contradiction.

Triple $(-1, 1, 2)$*:* By the symmetric argument, interchanging (A, t_1, t_3) with (B, t_2, t_4). □

Hence when $N = 2$: $S_4 \leq p - 1$ and $w = 2p - S_4 \geq p + 1$.

Steps 1–6 cover all cases, completing the proof of Proposition 6 and hence of Theorem 3. □

6. Consequences

Corollary 13. *For every odd prime p, $\mathsf{s}(C_p^3) \geq 9(p-1) + 1$.*

Proof. Adjoin the zero vector $\mathbf{0}$ to $\mathcal{A}_8$, giving nine vectors. The sequence $S' = \mathbf{0}^{p-1} \cdot S$ has length $9(p-1)$. A zero-sum subsequence of S' of length exactly p either involves some $\mathbf{a}_j \neq \mathbf{0}$—in which case the nonzero terms form a short zero-sum in S, contradicting Theorem 3—or consists entirely of copies of $\mathbf{0}$, which requires p copies, but only $p - 1$ are available. □

Remark 14. For $p = 3$, Theorem 3 gives $\eta(C_3^3) \geq 17$. Combined with the upper bound $\eta(C_3^3) \leq \mathsf{s}(C_3^3)$ and the known value $\mathsf{s}(C_3^3) = 19$ (see Edel, Elsholtz, Geroldinger, Kubertin, and Rackham [1]), this gives $17 \leq \eta(C_3^3) \leq 19$.

7. Remarks

7.1. **Computational verification.** The inequality $w(t) \geq p + 1$ can be verified by exhaustive enumeration over all p^5 parameter vectors. We have verified the bound for $p = 3$ (243 vectors), $p = 5$ (3,125 vectors), and $p = 7$ (16,807 vectors), confirming in each case that the minimum lift-weight is exactly $p+1$. The fact that the minimum is attained shows that the bound in Proposition 6 is tight: no additional element of $\mathbb{F}_p^3$ can be appended to $\mathcal{A}_8$ with multiplicity $p-1$ while preserving the zero-sum-free property.

7.2. **Machine-checked proof.** The proof of Proposition 6 has been formalized in Lean 4 with the Mathlib library [7]. The formalization works entirely in $\mathbb{Z}$ with explicit bounds on the parameters, avoiding modular arithmetic at the level of Lean types. The case analysis on the wrap count triple (n_0, n_1, n_2) is carried out by the `omega`, `linarith`, and `nlinarith` tactics, with the parity argument in the $N = 2$ case resolved by `omega` on the equation $2T = p$. The complete source file `DavenportAll.lean` (541 lines) accompanies this paper as ancillary material on the arXiv.

7.3. **Relation to Elsholtz's work.** Elsholtz [3] proved $\mathsf{s}(C_n^3) \geq 9(n-1) + 1$ for all odd n using nine vectors, one of which is the zero vector. His parity obstruction—deriving $2x = n$ for some integer x, contradicting oddness of n—is the same idea underlying our $N = 2$ case. The essential distinction is that his construction bounds s (zero-sums of length *exactly* n) while ours bounds η (zero-sums of length *at most* p), a strictly stronger condition requiring all vectors to be nonzero.

7.4. **Open problems.** For $p \geq 5$, the exact value of $\eta(C_p^3)$ remains unknown. Our lower bound $8(p-1)+1$ and the trivial upper bound $\eta(C_p^3) \leq \mathsf{s}(C_p^3)$ leave a gap. Determining whether $\eta(C_p^3) = 8(p-1) + 1$—equivalently, whether no set of nine nonzero vectors in $\mathbb{F}_p^3$ can be zero-sum-free for all lengths $1, \ldots, p$ at multiplicity $p - 1$—is the natural next question.

For rank $r \geq 4$, it is open whether the Harborth bound $\eta(C_p^r) \geq (2^r - 1)(p - 1) + 1$ can be improved.

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(To be added.)

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